Semantic Integration of Textual Data in Official Statistics:

Challenges and opportunities

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Non-traditional data sources

Modern **ML/AI methods** enable the integration of non-traditional data sources in the statistical production process.

- **Questionnaire text fields: semantic search methods** can support the assignment of classification codes to user queries (e.g., NACE, ATECO).
- **Enterprise websites:** web scraping combined with **AI-based retrieval approaches** enables the extraction of information on enterprises' products and services, and supports their classification (e.g., PRODCOM).
- **Patent data: embedding-based methods** can be used, for example, to estimate occupations' exposure to AI.

Automated analysis of Enterprise Websites

Starting in 2013, Istat began exploring the potential of web scraping techniques within both national and European context.

- **E-commerce detection:** establish whether an enterprise sells goods or services online.
- **Online job vacancies detection:** establish whether an enterprise advertises job openings on its website.

Statistical objective: use AI/ML methods to derive statistical indicators on firms' digitalisation.

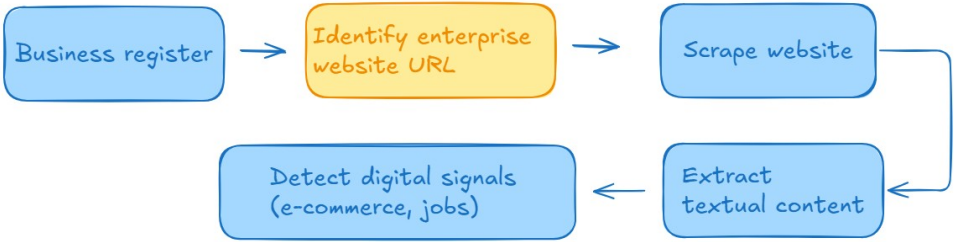
Collecting data from web sources

Web scraping involves identifying and extracting relevant information from enterprise websites. Key challenges include:

- identifying the correct website
- navigating complex and heterogeneous page structures
- extracting relevant content from noisy data

Web data is rich, but difficult to systematically collect and standardize.

Web scraping pipeline



Website identification is a critical step,
as it directly affects the quality of all downstream analyses.

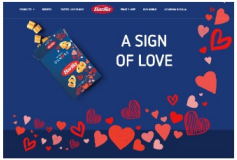
Website Identification: Model Performance

Model	Accuracy	Recall	F1	AUC
Logistic Regression	0.862	0.785	0.778	0.841
Gradient Boosting	0.867	0.813	0.790	0.852
Neural Network	**0.868**	**0.827**	**0.794**	**0.857**

The **neural network model** achieves the best performance across all metrics and is selected as the final model.

From web data to statistical codes

Access web sources



Extract product information

Assign Codes

SEZIONE PRODOTTI

PRODOTTI(*)		PRODUZIONE REALIZZATA
Codici prodotti *(scelti tra quelli elencati nel foglio Lista prodotti 23)*	**Unità di misura**	**Produzione realizzata - Quantità**
1	**2**	**6**
10.73.11	kg	11200
10.61.21	kg	12500
10.13.15	kg	7300
10.73.12	kg	1200

Web scraping provides **raw, unstructured data.**

The next step is transforming this information into statistical variables through automated coding.

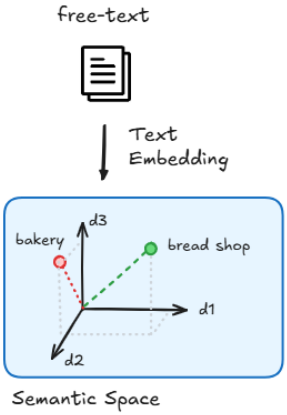
Semantic matching for automated coding

Product descriptions are compared with predefined statistical categories, using **numerical vectors**, called **embeddings**.

Using embeddings:

- both descriptions and categories are represented in the same semantic space
- similarity can be measured using **distance metrics** (e.g. cosine similarity)

This allows automated coding systems to match **meanings** rather than individual **words**



Assessing Reliability of AI-based coding

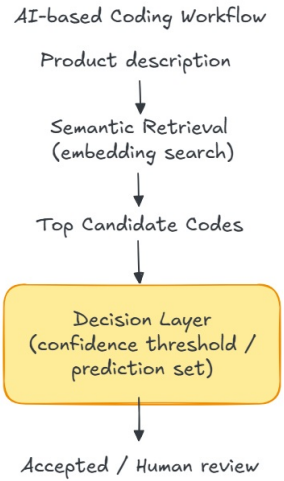
Semantic retrieval generates candidate codes.

How can we assess whether **the prediction can be considered reliable**?

Reliability can be assessed through:

- confidence thresholds
- prediction sets
- uncertainty quantification

Developing these methods requires **new statistical and computational skills** within statistical institutions.



AI in official statistics: opportunities and responsibilities

- **New opportunities for data integration** AI enables statistical institutions to integrate large volumes of unstructured textual data into official statistics.
- **Statistical reliability remains essential** AI outputs must be supported by validation, uncertainty quantification, and quality assessment.
- **New skills for statistical institutions** AI is transforming statistical production pipelines, requiring new methods, tools, and stronger AI literacy across statistical institutions.

The challenge is not only to use AI, but to use it responsibly in Official Statistics.

Istat

Q&A

Thank you for listening!

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